Bézout and integer equations

Advanced high school · extension · Lesson and method · 04

Prerequisites. Euclidean algorithm and divisibility in $\mathbb{Z}$.

Key point · Ideas and method. For positive integers a and b, the Euclidean algorithm computes $d = \gcd(a, b)$. Back-substitution expresses d as au + bv for integers u and v: a Bézout identity.
The equation ax + by = c has integer solutions exactly when d divides c. Necessity follows because d divides ax + by; sufficiency follows by multiplying a Bézout identity by $\frac{c}{d}$.
If (x_0, y_0) is one solution, all solutions are $x = x_0 + (\frac{b}{d})t$ and $y = y_0 - (\frac{a}{d})t$ for integer t. Conditions $x \geq 0$ and $y \geq 0$ must then be imposed separately.

Example 1. For $15x + 21y = 6$, use $21 = 15 + 6$ and $15 = 2 \times 6 + 3$. Thus $3 = 3 \times 15 - 2 \times 21$. One solution is $(6, -4)$, and all solutions are $x = 6 + 7t, y = -4 - 5t$.

Activity 1 · Explore. Try several values of t in the example and verify the equation exactly.